

1. Gradient Descent Method in "Mathematics":

A. gradient descent method in one dimension:

$$x_{n+1} = x_n + \eta \cdot (-f'(x_n))$$

$y = f(x)$ is function
 x_{n+1} is the next point moving from x_n .

B. gradient descent method in multidimension:

$$\begin{bmatrix} x'_0 \\ x'_1 \\ \vdots \\ x'_n \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} + \eta \cdot \begin{pmatrix} -\frac{\partial f}{\partial x_0} \\ -\frac{\partial f}{\partial x_1} \\ \vdots \\ -\frac{\partial f}{\partial x_n} \end{pmatrix} \longrightarrow \vec{x}' = \vec{x} + \eta \cdot (-\nabla f)$$

$y = f(x_0, x_1, \dots, x_n)$ is function

$\begin{bmatrix} x'_0 \\ x'_1 \\ \vdots \\ x'_n \end{bmatrix}$ is the next point moving from $\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$

$\vec{x}' = \begin{bmatrix} x'_0 \\ x'_1 \\ \vdots \\ x'_n \end{bmatrix}$, is a vector.

$\vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$, is a vector.

C. step size, $|\eta \cdot (-f'(x_n))|$ or $\left| \eta \cdot \begin{pmatrix} -\frac{\partial f}{\partial x_0} \\ -\frac{\partial f}{\partial x_1} \\ \vdots \\ -\frac{\partial f}{\partial x_n} \end{pmatrix} \right|$, depends on learning rate (η) and (partial) derivative $\begin{pmatrix} -f'(x_n) \text{ or } -\begin{bmatrix} \frac{\partial f}{\partial x_0} \\ \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \end{pmatrix}$,

so step size will decrease as size of (partial) derivative decrease.

as gradient descent method converge:

$y = f(x)$ or $y = f(x_0, x_1, \dots, x_n)$ function converge at minimum point(s) $\longrightarrow$

the size of (partial) derivative approach 0 $\longrightarrow$

which implies that step size also approach 0.

D. if learning rate (η is always a positive number) is set to too large value,

$y = f(x)$ or $y = f(x_0, x_1, \dots, x_n)$ function converge at minimum point(s) $\longrightarrow$

the size of (partial) derivative approach 0 $\longrightarrow$

but learning rate is too big for step size to approach 0. $\longrightarrow$

gradient descent method tend to overshoot the solution and fail to converge.

2. Optimizing Problem in Perceptron Learning Algorithm:

in perceptron learning algorithm, we hope "predicted output P_y " & "output y (ground truth)" can be the same finally.

on the other words, we define the error function: "Mean Square Error (MSE) $\frac{1}{m} \sum_{i=0}^{m-1} (y_i - \hat{y}_i)^2$ " as a standard indicator for m set of pair training examples,

y_i is output y from a pair training example
 $\hat{y}_i$ is predicted output P_y from training perceptron

to treat our optimizing problem whether we find out appropriate weights that minimize the value of error function ($\equiv y_i - \hat{y}_i$ approach 0) or not